# A quadrisection identity between OEIS A084703 and OEIS A089928

Adrian Perez Fontelles
Independent researcher

31 August 2026

**Abstract**

OEIS A084703 carries an empirical observation of A. Ratushnyak that its $n$-th term equals the $(4n-2)$-nd term of OEIS A089928. It is true, and the proof is a single line once both entries' own closed forms are put side by side: the exponent $4n$ produced by the quadrisection turns $(1 \pm \sqrt{2})^{4n}$ into $(17 \pm 12\sqrt{2})^n$, and the parity term $(-1)^{\lfloor (4n-2)/2 \rfloor}$ is constantly $-1$. No generating function is needed, and in particular the generating function recorded on A089928 is not used — as noted below, it does not reproduce that entry's own terms.

2020 Mathematics Subject Classification. 11B37, 11B39, 11D09.

## 1 The two sequences and the conjecture

OEIS A084703 is "Squares k such that 2*k+1 is also a square.". It has offset 0 and begins

$$0, 4, 144, 4900, 166464, 5654884, 192099600, \ \ldots$$

OEIS A089928 is "a(n) = 2*a(n-1) + 2*a(n-3) + a(n-4), with a(0)=1, a(1)=2, a(3)=4, a(4)=10.". It has offset 0 and begins

$$1, 2, 4, 10, 25, 60, 144, 348, \ \ldots$$

The entry A084703 carries the following comment:

> Empirical: a(n) = A089928(4*n-2), for n > 0. — *Alex Ratushnyak*, Apr 12 2013

As of the "Last modified" line on the live entry (Jan 05 2026 20:36:39, revision 68) the statement is still recorded as unproved, and nothing on either entry records it as settled either way.

Written out, the statement to be proved is

$$a(n) \ = \ b(4n-2) \qquad \text{for all } n \geq 1, \tag{1}$$

where $a$ is A084703 and $b$ is A089928.

## 2 What each entry states as fact

A084703 states, not as a conjecture,

> a(n) = ((17+12*sqrt(2))^n + (17-12*sqrt(2))^n - 2)/8.

that is,

$$a(n) \ = \ \frac{\left(17+12\sqrt{2}\right)^n + \left(17-12\sqrt{2}\right)^n - 2}{8}. \tag{2}$$

A089928 states, not as a conjecture,

a(n) = ( (1+sqrt(2))^(n+2) + (1-sqrt(2))^(n+2) + 2*(-1)^floor(n/2) )/8.

that is,

$$b(m) \;=\; \frac{\left(1+\sqrt{2}\right)^{m+2} + \left(1-\sqrt{2}\right)^{m+2} + 2(-1)^{\lfloor m/2 \rfloor}}{8}. \tag{3}$$

Each was checked against its own entry's DATA at every published index before use: (2) against all 18 terms of A084703 and (3) against all 30 terms of A089928, in exact arithmetic in $\mathbb{Z}[\sqrt{2}]$. These two lines are the only input taken from the entries.

**Remark 1.** A089928 also carries the line "G.f.: 1/((1+2*x)*(1-2*x-x^2))". That is not the generating function of A089928: the stated denominator expands to $1-5x^2-2x^3$, whose reciprocal begins $1, 0, 5, \ldots$, whereas the entry begins $1, 2, 4, \ldots$. The factor $(1+2x)$ should read $(1+x^2)$, since $(1-2x-x^2)(1+x^2) = 1-2x-2x^3-x^4$, matching the recurrence in the entry's name. Nothing below uses that line; it is recorded only so that a reader checking the entry is not misled.

## 3 The proof

**Lemma 2.** $\left(1+\sqrt{2}\right)^4 = 17+12\sqrt{2}$ *and* $\left(1-\sqrt{2}\right)^4 = 17-12\sqrt{2}$.

*Proof.* $\left(1\pm\sqrt{2}\right)^2 = 1\pm 2\sqrt{2}+2 = 3\pm 2\sqrt{2}$, and $\left(3\pm 2\sqrt{2}\right)^2 = 9\pm 12\sqrt{2}+8 = 17\pm 12\sqrt{2}$. □

**Lemma 3.** *For every integer* $n \geq 1$, $(-1)^{\lfloor (4n-2)/2 \rfloor} = -1$.

*Proof.* $4n-2$ is even, so $\lfloor (4n-2)/2 \rfloor = 2n-1$, which is odd. □

**Theorem 4.** *For every integer* $n \geq 1$, $a(n) = b(4n-2)$, *where $a$ is OEIS A084703 and $b$ is OEIS A089928. This is the statement of Section 1.*

*Proof.* Put $m = 4n-2$ in (3). Then $m+2 = 4n$, so by Lemma 3

$$b(4n-2) = \frac{\left(1+\sqrt{2}\right)^{4n} + \left(1-\sqrt{2}\right)^{4n} - 2}{8}.$$

By Lemma 2, $\left(1\pm\sqrt{2}\right)^{4n} = \left(\left(1\pm\sqrt{2}\right)^4\right)^n = \left(17\pm 12\sqrt{2}\right)^n$, so

$$b(4n-2) = \frac{\left(17+12\sqrt{2}\right)^n + \left(17-12\sqrt{2}\right)^n - 2}{8},$$

which is exactly $a(n)$ by (2). □

**Remark 5.** The identity in fact holds at $n = 0$ as well: both sides are 0, since $b(-2)$ is not defined by the entry but (3) evaluated at $m = -2$ gives $(1+1-2)/8 = 0 = a(0)$. The restriction $n > 0$ in the comment is therefore only a statement about which indices of A089928 are published.

## 4 Verification

Three checks, each independent of the algebra above.

First, formula (2) was evaluated in exact arithmetic in $\mathbb{Z}[\sqrt{2}]$ and compared against every one of the 18 published terms of A084703; the values agree exactly. A formula that reproduced only some of them would not have been used.

Second, formula (3) was evaluated the same way and compared against every one of the 30 published terms of A089928; the values agree exactly. It was also checked against the recurrence $b(m) = 2b(m-1) + 2b(m-3) + b(m-4)$ named on that entry.

Third, the identity (1) was checked directly on the published terms, without either formula: for every $n$ with $1 \le n \le 7$ — the range for which both $a(n)$ and $b(4n-2)$ are published — the two integers agree.

## References


[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026.

[2] G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, 6th ed., Oxford University Press, 2008.

[3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011.